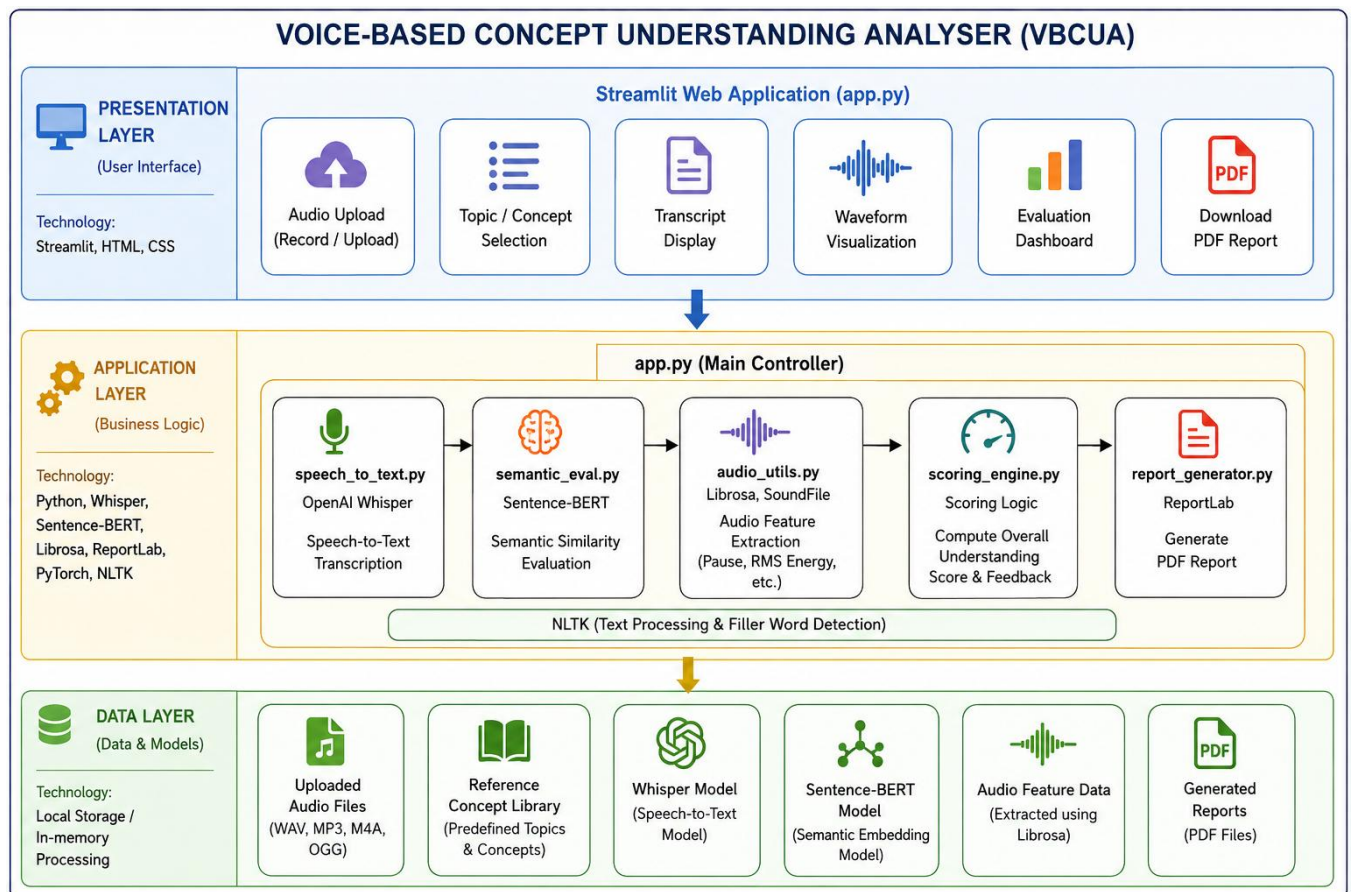


Solution Architecture

Date	29 June 2026
TeamID	XXXXXX
ProjectName	Voice Based Concept Understanding Analyser
Maximum Marks	5 Marks

Solution Architecture Diagram:

Provide a high-level visual representation of your solution's architecture. Use the structural layout below as a template to map out your Presentation Layer, Application Layer, and Data Layer. Fill in the specific technologies and components for your project directly into the boxes to create your architectural diagram.



Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
Presentation Layer	Provides an interactive web interface where users upload audio recordings, select a concept, view transcripts, evaluation metrics, waveform visualizations, and download PDF reports.	Streamlit, HTML, CSS
Application Layer	Receives uploaded audio, validates input, coordinates speech transcription, semantic similarity analysis, audio feature extraction, understanding score calculation, and report generation.	Python, Streamlit
Core Logic Service	Converts speech to text, evaluates conceptual understanding, extracts audio features, computes the final understanding score, and generates qualitative feedback.	OpenAI Whisper, Sentence-BERT, Librosa, PyTorch, NLTK, ReportLab
External AI Integrations	Integrates AI models for speech transcription and semantic similarity. Future versions may support cloud AI services, multilingual models, and LMS integration.	OpenAI Whisper, Sentence-Transformers (Sentence-BERT) (Future: Google Speech API, LMS APIs)
Data / Storage Layer	Stores uploaded audio temporarily, reference concept library, AI models, extracted audio features, and generated PDF evaluation reports. The current version performs processing without a dedicated database.	Local File Storage, Whisper Model, Sentence-BERT Model, PDF Files (ReportLab)